Missense Impact: Population Aware Genomic Models for Missense Variant Effect Prediction

Literature Review & Elaboration of Problem

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Version: 02.30

Project Code	F25-68
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Submission Date	4 December 2025

Document History

Version Name of Person	*Date*	*Description of change*
Sami Ur Rehman	10/20/25	Document Created
Bilal Asif Burney	10/29/25	Added initial Literature Review draft; included reference list (IEEE format)
Sami-Ur-Rehman	11/4/25	Expanded Background & Justification section with additional citations
Sami-Ur-Rehman	11/10/25	Added refined Literature Review focusing on ancestry bias & recalibration
Mustafa Waqar	11/14/25	Added Methodology section (bias quantification, Bayesian recalibration, PLM domain adaptation)
Mustafa Waqar	11/20/25	Updated Clinical Pipeline section aligned with ACMG/AMP guidelines
Bilal Asif Burney, Sami Ur Rehman	11/27/25	Final edits for clarity, formatting, and technical validation
Sami Ur Rehman, Mustafa Waqar, Bilal Asif Burney	12/4/25	Finalized edit. (Overall Review)

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Version	*Sign-off Authority*	*Project Role*	*Sign-off Date*
1.50	Dr. Nasir Uddin	Supervisor	11/10/25
1.70	Dr. Nasir Uddin	Supervisor	11/14/25
2.00	Dr. Nasir Uddin	Supervisor	11/20/25
2.10	Dr. Nasir Uddin	Supervisor	11/27/25
2.20	Dr. Nasir Uddin	Supervisor	12/02/25
2.30	Dr. Nasir Uddin	Supervisor	12/04/25

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1. Abstract

Missense variants, caused by single amino acid substitutions, are among the most common forms of genetic variation and are central to understanding monogenic diseases, cancer susceptibility, and complex traits. Although many computational tools exist to assess their pathogenicity, current prediction systems consistently underperform on South Asian genomes due to limited representation in reference databases and model training sets. This leads to inflated pathogenicity scores, poor calibration, and a disproportionately high number of variants classified as uncertain.

The core objective of this research is to address these disparities by developing a South Asian calibrated pathogenicity prediction framework. The proposed work focuses on identifying and quantifying biases in widely used classical, ensemble, generative, and protein language model predictors when applied to South Asian genomes. By systematically analysing ancestry-linked performance gaps, the project establishes the foundation for region-specific recalibration strategies.

To correct these biases, the methodology integrates population-specific allele-frequency information from major South Asian genomic resources and incorporates it into recalibration layers built using Bayesian, logistic, and domain-adaptation techniques. These adapted models will be evaluated through ancestry-stratified benchmarking to ensure improved accuracy, reliability, and fairness across diverse South Asian subpopulations. The framework will culminate in a standardized clinical annotation pipeline aligned with established variant interpretation guidelines.

The expected outcome of this research is a suite of calibrated prediction models, an ancestry-aware evaluation environment, and a deployable clinical workflow tailored to South Asian genetic data. By enhancing predictive reliability and reducing interpretation uncertainty, this work contributes both scientifically and ethically to equitable genomic medicine and supports more accurate diagnosis for one of the world's most underrepresented populations.

2. Background and Justification

Missense variant interpretation has traditionally depended on computational tools that leverage evolutionary conservation and protein sequence structure features. Classical predictors such as PolyPhen-2 [1] and SIFT [2] estimate the functional impact of amino acid substitutions by analyzing evolutionary constraint across aligned homologous sequences. While these approaches perform well in deeply conserved regions, they show reduced reliability for positions with limited alignment depth, lineage-specific variation, or low-quality evolutionary signal. Ensemble-based methods such as REVEL [3] attempt to mitigate these limitations by integrating information from multiple models, yet they remain constrained by the biases and inconsistencies present in their upstream tools and training labels.

Recent advances in deep learning have transformed variant effect prediction by enabling models to capture higher-order biochemical and structural relationships. Evolutionary generative frameworks such as EVE [9] model sequence variation through deep probabilistic learning, while large-scale protein language and structural models such as AlphaMissense [11] use representations learned from millions of protein sequences and predicted three-dimensional conformations. Although these modern methods achieve superior predictive performance compared to classical algorithms, they still lack direct incorporation of human allele-frequency information. This limitation is significant because a variant that appears evolutionarily unusual may nonetheless be

common and benign in human populations, particularly in groups with distinct demographic histories.

An expanding body of evidence demonstrates that genomic resources including ClinVar, HGMD, and gnomAD are disproportionately composed of individuals of European ancestry. This imbalance introduces systematic ancestry-dependent performance disparities across variant effect predictors. Studies consistently show inflated pathogenicity estimates, poor score calibration, and elevated rates of variant misclassification for underrepresented populations. Zhou et al. reported sharply reduced sensitivity for multiple tools when evaluated across diverse ancestries [4]. Lee et al. demonstrated that population-mismatched filtering leads to substantial increases in false positives in Asian cohorts [6]. Sharo et al. found extensive misclassification of South Asian variants across ClinVar and HGMD [7]. Additionally, work inspired by population-aware generative frameworks such as popEVE [12] highlights that incorporating human allele frequencies directly into predictive models improves calibration and reduces misinterpretation.

South Asians represent roughly one-quarter of the global population yet remain among the most underrepresented groups across all major genomic databases. This underrepresentation results in limited labelled training data, incomplete allele-frequency coverage, and an absence of models calibrated to the genetic structure of South Asian populations. Therefore, current predictors frequently misclassify common South Asian variants as pathogenic or label them as Variants of Uncertain Significance (VUS), creating significant barriers to accurate clinical diagnosis.

The need for this research is therefore justified on scientific, methodological, and ethical grounds. Scientifically, improved modelling is required to capture South Asian population structure, genetic diversity, and variant frequency distributions accurately. Methodologically, recalibration and adaptation of state-of-the-art tools are necessary to ensure reliable predictions across globally diverse genomes. Ethically, population-inclusive computational genomics is essential for equitable healthcare, particularly as genomic medicine becomes increasingly integrated into clinical practice. By addressing these gaps, this research advances the development of population-aware variant effect prediction and supports more accurate and fair genetic interpretation for South Asian communities.

3. Problem Statement

Missense variant pathogenicity predictors exhibit systematic ancestry bias that results in misclassified, mis-calibrated, or uncertain variant interpretations for South Asian genomes. Because most existing computational tools are trained, calibrated, and benchmarked on datasets enriched for European individuals, they tend to overestimate the pathogenicity of variants that are common and benign within South Asian populations. This leads to inflated posterior probabilities, unreliable risk predictions, and disproportionately high rates of Variants of Uncertain Significance (VUS), impairing clinical decision-making and precision medicine outcomes for two billion South Asians.

3.1 Core Problem

- ***Lack of South Asian allele-frequency priors:***
  Widely used predictors, including PolyPhen-2, SIFT, REVEL, EVE, and AlphaMissense, do not incorporate South Asian specific allele frequency information. As a result, their scores fail to reflect true population-level pathogenicity likelihoods and systematically misclassify benign variation as potentially harmful.

- ***Eurocentric training and reference data:***
  Clinical databases such as ClinVar and HGMD, along with population reference resources like gnomAD, are heavily skewed toward European ancestries. When applied to South Asian genomes, predictors trained on these skewed datasets produce biased outputs, leading to calibration errors, false positives, and unreliable clinical interpretations.

3.2 Associated Challenges

- ***Limited labelled South Asian variants:***
  The scarcity of South Asian validated pathogenic and benign variants limits the effectiveness of supervised model fine-tuning or traditional performance calibration.

- ***High VUS burden:***
  South Asian patients experience disproportionately high rates of VUS calls due to poor predictor calibration and limited representation in clinical datasets, complicating clinical decision-making.

- ***Calibration and ACMG alignment issues:***
  Ancestry-specific calibration disparities reduce the reliability of mapping predictor outputs to ACMG/AMP evidence criteria (e.g., BA1/BS1, PP3/BP4), leading to inconsistent or incorrect variant classifications.

- ***Dataset heterogeneity:***
  Differences across gnomAD-SAS, GenomeAsia100K, IndiGen, and related regional resources, including sample sizes, sequencing technologies, and population substructure, pose challenges for constructing unified South Asian priors and harmonized benchmarking datasets.

3.3 Research Need

To address these issues, there is a critical need for a comprehensive, population-aware recalibration and model adaptation pipeline tailored specifically to South Asian genomic datasets. Such a framework must integrate South Asian allele frequency priors, correct ancestry-dependent calibration discrepancies, adapt modern generative and protein language models to South Asian population structure, and ensure dependable, clinically aligned pathogenicity estimates. This work is essential for improving fairness, accuracy, and utility in genomic interpretation for South Asian individuals.

4. Literature Review

4.1 Classical and Ensemble Variant Effect Predictors (Clinical-Trained)

Early computational interpretation relied on "Clinical-Trained" models that used supervised learning on labelled datasets such as ClinVar. Classical tools such as PolyPhen-2 [2] and SIFT [1] laid the foundation by estimating functional impact through evolutionary conservation sequence alignment. To improve accuracy, ensemble methods like REVEL [3] integrate scores from multiple tools. However, these models suffer from data circularity (Type 1 and Type 2), where variants used to train the model are also used to evaluate it, artificially inflating performance metrics. Critically, because their training data (ClinVar and HGMD) [7] is over 70% European, these models learn to recognize European benign variants while misclassifying equally benign South Asian variants as "Variants of Uncertain Significance" (VUS) or pathogenic.

4.2 Evolutionary and Generative Models (Population-Free)

To circumvent clinical ascertainment bias, "Population-Free" models were developed. EVE (Evolutionary model of Variant Effect) [9] uses deep variational autoencoders (VAEs) trained solely on evolutionary sequence alignments, without seeing human clinical labels. Similarly, Protein Language Models (PLMs) like ESM-1b and ProtBERT learn the "grammar" of protein sequences from millions of unlabelled peptides. Recent benchmarking demonstrates that these population-free models exhibit the lowest ancestry bias because they are blind to skewed human clinical data. However, evolutionary novelty in human populations is not equivalent to pathogenicity: variants that appear evolutionarily “rare” may be frequent and benign in specific ancestries. Without population-specific allele frequency integration, population-free models can systematically penalize variants prevalent in South Asian groups.

4.3 Structure-Aware and Population-Tuned Models

The current state-of-the-art is represented by AlphaMissense [11], which combines PLM embeddings with AlphaFold structural context. Unlike pure PLMs, AlphaMissense is "Population-Tuned": it uses human allele frequency data as a weak supervision signal. While this improves general performance, it re-introduces bias. If the allele frequency reference is Eurocentric (e.g., gnomAD-EUR), the model may over-predict pathogenicity for South Asian variants that are absent in European cohorts but common locally.

4.4 The Confounding Role of Intrinsic Disorder

A critical but often overlooked factor in VEP performance is Intrinsic Protein Disorder. Recent analyses reveal that VEPs perform artificially well on disordered protein regions because these regions are evolutionarily variable and enriched for benign variants. This inflation masks inferior performance in structured, disease-relevant domains. An accurate evaluation framework for South Asian genomes must therefore stratify results by protein structure (Ordered vs. Disordered) to avoid misleading accuracy metrics.

4.5 Population-Aware Prediction and Calibration

Efforts such as PopEVE [12] demonstrate that incorporating human allele frequency data into generative frameworks improves calibration and reduces systematic misclassification. By integrating frequency information from diverse populations, PopEVE aligns evolutionary likelihood scores with actual human variation, highlighting an effective path toward population-aware recalibration. These findings support the feasibility of adapting modern predictors to South Asian cohorts through Bayesian reweighting and logistic score adjustment.

4.6 Evidence of Ancestry Bias

Multiple studies provide direct evidence of ancestry-linked performance disparities. Zhou et al. [4] identify significant sensitivity drops in popular predictors when evaluating African and Asian genomes. Lee et al. [6] show that population-matched allele frequency filtering dramatically reduces false positives in Asian cancer datasets, underscoring the consequences of population mismatch. Sharo et al. [7] reveal widespread inconsistencies between pathogenic labels and real-world population frequencies, with South Asian variants disproportionately affected. Collectively, these studies highlight the urgent need for population-specific adaptation and recalibration.

4.7 Proposed Methodology and System Architecture

This project adopts a structured five-stage methodology aimed at quantifying, mitigating, and validating ancestry-dependent bias in state-of-the-art missense Variant Effect Predictors (VEPs), with a specific emphasis on South Asian (SAS) genomic cohorts. The workflow proceeds from data collection to clinical-grade deployment, ensuring fairness, reproducibility, and alignment with ACMG/AMP variant interpretation guidelines.

4.7.1 Stage 1 -> Data Collection and Curation

This stage focuses on constructing a high-quality, ancestry-resolved genomic dataset representative of South Asian genetic diversity.

4.7.1.1 Data Sources

Primary Population Datasets

- GenomeAsia100K (~5,734 SAS genomes) and the gnomAD v4.1 SAS subset serve as the main sources of allele-frequency information. These datasets include substantial numbers of novel SAS variants absent from global cohorts.
- IndiGen (1,029 whole genomes) and the SAS super-population of 1000 Genomes provide additional population-stratified variants.

Validation Cohorts

- A locally sourced AKU Pakistani exome dataset (~2,000 samples) is used as an external validation and robustness-checking cohort.

Clinical Ground-Truth Labels

- Pathogenic and benign assertions are obtained from ClinVar, filtered for high-confidence submissions to reduce annotation noise.

4.7.1.2 Preprocessing Pipeline

Normalization

- All VCF files undergo left-alignment, decomposition, and normalization using *bcftools norm*, ensuring consistent representation of multi-allelic and indel variants.

Functional Annotation

- Variants are annotated using Ensembl VEP with the --canonical flag to resolve transcript ambiguity and correctly map genomic coordinates to protein positions.

Sequence Construction

- Full-length wild-type and mutant protein sequences are generated to support downstream Protein Language Model (PLM) fine-tuning.

Outcome

- A harmonized, population-stratified variant dataset suitable for bias assessment and recalibration.

4.7.2 Stage 2 -> Bias Quantification Framework

This stage characterizes the baseline ancestry-dependent bias embedded in contemporary VEPs.

4.7.2.1 Predictors Evaluated

The analysis covers representative categories of pathogenicity predictors:

- Clinical-trained models: REVEL
- Ensemble predictors: Meta-models integrating multiple features
- Evolutionary models: EVE, popEVE
- Protein Language Models: ESM-1b, ProtBERT

4.7.2.2 Bias Metrics

Sensitivity Drop

- Measures the difference in pathogenic variant sensitivity between ancestral groups.

False Positive Rate (Population Mismatch)

- Evaluates erroneous pathogenic calls on variants common in SAS populations (AF > 1%) but rare globally.

Stratified Testing

- A balanced, ancestry-labelled subset of ClinVar variants is used to quantify cross-population drift.

Outcome

- A statistically validated bias profile establishing the need and direction for recalibration.

4.7.3 Stage 3 -> Model Adaptation and Recalibration

This stage implements the core technical contribution: integrating population-specific priors and fine-tuning models for improved fairness.

4.7.3.1 Bayesian Recalibration

A Bayesian correction layer modifies raw VEP scores (S) by incorporating SAS-specific allele frequency priors:

- Prior terms distinguish common SAS variants from globally rare ones, correcting systematic pathogenicity inflation.

4.7.3.2 Logistic Recalibration

Models outputting non-probabilistic scores (e.g., REVEL, popEVE) are mapped to calibrated probabilities via logistic regression trained exclusively on SAS-labelled variants.

4.7.3.3 PLM Fine-Tuning

Protein Language Models (EVE, ESM-1b, ProtBERT) are adapted using:

- Paired wild-type and mutant sequences.
- Domain-Adversarial Training to discourage ancestry-linked batch effects.

4.7.3.4 Ensemble Prediction Integration

Final predictions combine:

- Raw PLM embeddings
- Bayesian/logistic-corrected scores
- SAS population-frequency features via a weighted or stacking ensemble framework.

4.7.4 Stage 4 -> Evaluation and Benchmarking Strategy

A multi-dimensional evaluation framework assesses discrimination, calibration, fairness, and clinical utility.

4.7.4.1 Gene-Level Holdout Splitting

An 80/10/10 gene-level split prevents data leakage and ensures predictors cannot memorize homologous sequences.

4.7.4.2 Stratified Benchmarking

Results are reported across:

- Allele frequency bins: rare (AF < 0.001), low-frequency, and common (AF ≥ 0.01).
- Structural contexts: ordered regions (pLDDT > 50) and intrinsically disordered regions (pLDDT < 50).

4.7.4.3 Fairness Metrics

Bias reduction is measured using the Equalized Odds Gap, representing differences in true-positive and false-positive rates across ancestries.

- A reduction ≥ 25% is deemed a significant fairness improvement.

4.7.4.4 Clinical Performance Metrics

Evaluation includes:

- AUROC, AUPRC
- Calibration slope, Brier score
- Decision-curve analysis
- Net Reclassification Improvement (NRI)

4.7.5 Stage 5 -> Clinical-Readiness Protocol

This final stage prepares the recalibrated models for integration into clinical variant interpretation pipelines.

4.7.5.1 ACMG/AMP Tier Mapping

Calibrated probabilities are mapped to standardized ACMG/AMP categories:

- Benign
- Likely Benign
- VUS
- Likely Pathogenic
- Pathogenic

4.7.5.2 Bias Audit Generation

Per-batch audit logs include:

- Ancestry-stratified accuracy
- Calibration diagnostics
- Fairness alerts for high-risk variants

4.7.5.3 Reproducibility and Deployment

- Fully containerized environment (Docker/Singularity)
- Version-controlled pipelines via Jupyter/Colab
- Designed for external replication across independent datasets

5. Appendices

5.1 Models Used in Missense Pathogenicity Prediction

a. Classical Predictors

i. PolyPhen-2

- A supervised classifier that evaluates missense variants using:
  - Evolutionary conservation scores
  - Protein structural features
  - Functional domain context
- **Known weakness:** MSA depth and Eurocentric training labels indirectly influence its thresholds.

ii. ***SIFT***
- Predicts residue tolerance using conservation in multiple sequence alignments (MSAs).
- Variants at positions with low evolutionary tolerance are predicted to be damaging.
- SIFT does not incorporate human allele-frequency information.

b. Ensemble Predictors

i. ***REVEL***
- An ensemble meta-classifier integrating 13+ features, including SIFT, PolyPhen-2, and conservation metrics.
- **Strength:** high overall accuracy
- **Bias source:** training labels from ClinVar + HGMD ⇒ Eurocentric

c. Evolutionary Generative Models

i. ***EVE***
- An unsupervised variational autoencoder (VAE) trained on protein MSAs to learn evolutionary constraints.
- **Strengths:** label-free, avoids clinical bias.
- **Limitation:** does not incorporate human allele frequency, which is essential for population-aware predictions.

ii. ***popEVE***
- Extends EVE by incorporating human population variation (presence/absence in cohorts).
- **Advantages:**
  - Better calibration for human datasets
  - More reliable distinction between benign polymorphisms and rare pathogenic variants

d. Protein Language Models (PLMs)

i. ***ESM-1b and ESM-2***
- Transformer-based models pretrained on millions of protein sequences (UniRef).
- They produce contextual embeddings for each residue and variant.
- **Properties:**
  - No clinical labels
  - No human allele frequency information
  - Sequence only modelling.
- Useful for capturing biochemical and structural constraints.

ii. ***ProtBERT***
- A bidirectional transformer model trained using the BERT objective on large-scale protein databases.
- **Benefits:**
  - Good residue-level contextual understanding
  - Impressive performance when combined with downstream calibrators.

e. Structure-Aware Models

i. ***AlphaMissense***
- A deep learning model integrating:
  - Protein language model embeddings
  - AlphaFold2 structural context
  - Missense substitution modeling

5.2 Genomic Datasets

a. Population Genomic Cohorts

ii. ***1000 Genomes Project***

- **Format/Size:** VCF (~88M variants; ~2,500 individuals)
- **SAS Representation:** ~500 individuals (BEB, GIH, ITU, PJL, STU) Used for baseline South Asian allele frequencies.

iii. ***GenomeAsia 100K (2023 Expansion)***

- **Format:** VCF; 5,734 South Asian genomes
- **Coverage:** India, Pakistan, Bangladesh, Sri Lanka, Nepal ~23% novel variants; strong regional diversity.

iv. ***GenomeIndia Project***

- **Format:** VCF; ~10,074 individuals
- **Coverage:** 83 Indian ethnic groups. Provides extensive within-South-Asia variability.

v. ***IndiGenomes***

- **Format:** VCF (~55M variants; 1,029 genomes)
- **Representation:** Indian populations. Useful for regional allele-frequency granularity.

vi. ***SG10K Pilot***

- **Format:** WGS (4,810 individuals)
- **Representation:** SAS diaspora subgroups. Adds cross-regional diversity to SAS allele frequencies.

vii. ***Pakistani WGS (Riazuddin et al.)***

- **Format:** FASTQ (~318 GB; 4 high-coverage genomes)
- **Representation:** Pakistani individuals. Useful for model evaluation.

b. Clinical and Reference Datasets

viii. ClinVar

- **Format:** VCF (~2M variants). Contains clinical pathogenic/benign labels but lacks population annotation. Known to exhibit European submission bias.

ix. gnomAD

- **Format:** VCF (~76k genomes/exomes)
- **SAS Representation:** Underrepresented relative to European cohorts Provides population allele frequencies, but with a limited SAS sample size.

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